Q1.

```
SELECT specialty,
       COUNT(*) AS doctor_count,
       ROUND(AVG(salary), 2) AS avg_salary,
       SUM(salary) AS total_salary
FROM doctors
GROUP BY specialty
HAVING AVG(salary) > 170000
ORDER BY avg_salary DESC;
```

Q2.

```
SELECT d.doctor_name,
       d.specialty,
       s.doctor_name AS supervisor_name
FROM doctors d
LEFT JOIN doctors s ON d.supervisor_id = s.doctor_id
ORDER BY d.doctor_id;
```

Q3.

```
SELECT d.doctor_name,
       d.specialty,
       COUNT(DISTINCT p.patient_id) AS patient_count,
       COALESCE(SUM(t.cost), 0) AS total_treatment_cost
FROM doctors d
LEFT JOIN patients p ON p.doctor_id = d.doctor_id
LEFT JOIN treatments t ON t.patient_id = p.patient_id
GROUP BY d.doctor_id, d.doctor_name, d.specialty
ORDER BY d.doctor_id;
```

Q4.

```
SELECT DISTINCT p.patient_name,
       p.ward,
       p.age
FROM patients p
JOIN treatments t ON t.patient_id = p.patient_id
WHERE t.cost > 50000
ORDER BY p.patient_name;
```

Q5.

```
SELECT d.doctor_name,
       d.specialty,
       d.salary,
       sub.avg_sal AS specialty_avg
FROM doctors d
JOIN (
    SELECT specialty, ROUND(AVG(salary), 2) AS avg_sal
    FROM doctors
    GROUP BY specialty
) sub ON sub.specialty = d.specialty
WHERE d.salary > sub.avg_sal
```

```
ORDER BY d.specialty, d.salary DESC;
```

Q6.

```
SELECT doctor_name,
       specialty,
       salary,
       RANK() OVER (PARTITION BY specialty ORDER BY salary DESC) AS sal_rank
FROM doctors
ORDER BY specialty, sal_rank;
```

Q7.

```
SELECT p.patient_name,
       t.treatment,
       t.treat_date,
       t.cost,
       SUM(t.cost) OVER (
           PARTITION BY t.patient_id
           ORDER BY t.treat_date, t.treat_id
       ) AS running_cost
FROM patients p
JOIN treatments t ON t.patient_id = p.patient_id
ORDER BY p.patient_name, t.treat_date;
```